

## Mark schemes

### Q1.

(a)  $\int (t-1) \ln t \, dt$

$$u = \ln t \quad \frac{dv}{dt} = t - 1$$

*Differentiate + integrate, correct direction*

M1

$$\frac{du}{dt} = \frac{1}{t} \quad v = \frac{t^2}{2} - t$$

*All correct*

A1

$$\begin{aligned} \int &= \left( \frac{t^2}{2} - t \right) \ln t - \left( \frac{t^2}{2} - t \right) \times \frac{1}{t} (dt) \\ &= \left( \frac{t^2}{2} - t \right) \ln t - \int \left( \frac{t}{2} - 1 \right) (dt) \end{aligned}$$

*Condone missing brackets*

A1

$$= \left( \frac{t^2}{2} - t \right) \ln t - \frac{t^2}{4} + t (+c)$$

CAO

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A1

4

### Alternative

$$\int (t-1) \ln t$$

$$u = \ln t \quad v' = (t-1)$$

(M1)

$$u' = \frac{1}{t} \quad v = \frac{(t-1)^2}{2}$$

(A1)

$$\begin{aligned}
 \int &= \frac{(t-1)^2}{2} \ln t - \int \frac{(t-1)^2}{t} \frac{1}{t} dt \\
 &= \frac{(t-1)^2}{2} \ln t - \frac{1}{2} \int \frac{t^2 - 2t + 1}{t} dt \\
 &= \frac{(t-1)^2}{2} \ln t - \frac{1}{2} \int \left( t - 2 + \frac{1}{t} \right) dt
 \end{aligned}$$

(A1)

$$\frac{(t-1)^2}{2} \ln t - \frac{1}{2} \left[ \frac{t^2}{2} - 2t + \ln t \right]$$

(A1)

$$\begin{aligned}
 &= \frac{t^2}{2} \ln t - t \ln t + \frac{1}{2} \ln t - \frac{t^2}{4} + t - \frac{1}{2} \ln t \\
 &= \left( \frac{t^2}{2} - t \right) \ln t - \frac{1}{4} t^2 + t + c
 \end{aligned}$$

(4)

(b)  $t = 2x + 1$   
 $dt = 2 dx$  (RHS)

$$\frac{dt}{dx} = 2 \quad (LHS)$$

M1

$2x = t - 1,$

OE

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m1

$$\int = \int 2(t-1) \ln t \frac{dt}{2}$$

AG

A1

3

(c)  $[x]_b^1 = [t]_1^3$

Limit becoming 3

M1

$$\begin{aligned}
 \int &= \left[ \left( \frac{t^2}{2} - t \right) \ln t - \frac{t^2}{4} + t \right]_1^3 \\
 &= \left[ \left( \frac{9}{2} - 3 \right) \ln 3 - \frac{9}{4} + 3 \right] - \left[ 0 - \frac{1}{4} + 1 \right]
 \end{aligned}$$

Correctly sub. 1, 3 into their (a)

m1

$$= \frac{3}{2} \ln 3$$

CSO

A1

3

or

$$\int = \left[ \left( \frac{(2x+1)^2}{2} - (2x+1) \right) \ln (2x+1) - \frac{(2x+1)^2}{4} + (2x+1) \right]_0^1$$

Condone 1 slip

(M1)

$$= \left( \left( \frac{9}{2} - 3 \right) \ln 3 - \frac{9}{4} + 3 \right) - \left( 0 - \frac{1}{4} + 1 \right)$$

Correctly sub. 0, 1

(m1)

$$= \frac{3}{2} \ln 3$$

CSO

(A1)

(3)

[10]

**Q2.**

(a) (i)  $p(-1) = -1 + 7 - 6$

Finding  $p(-1)$

M1

$= 0$  therefore  $x + 1$  is a factor

Shown to  $= 0$  **plus statement**

A1

(ii)  $p(x) = (x + 1)(x^2 - x - 6)$

*Long division/inspection (2 terms correct)*

M1

*Quadratic factor correct*

A1

$p(x) = (x + 1)(x + 2)(x - 3)$

*May earn M1, A1 For correct second factor  
then A1 For  $(x + 1)(x + 2)(x - 3)$*

A1

3

(b) (i)  $A(-2, 0)$

*Condone  $x = -2$*

B1

1

(ii)  $\frac{x^4}{4} - \frac{7x^2}{2} - 6x (+c)$

*One term correct*

M1

*Another term correct*

A1

*(may have + c or not)*

*All correct unsimplified*

A1

$$\left[ \frac{81}{4} - \frac{63}{2} - 18 \right] - \left[ \frac{1}{4} - \frac{7}{2} + 6 \right]$$

*F(3) - F(-1) attempted in correct order*

m1

$= -32$

*CSO; OE*

A1

5

(iii) Area of shaded region = 32

*FT their (b)(ii) but positive value needed*

B1ft

1

(iv)  $\frac{dy}{dx} = 3x^2 - 7$

*One term correct*

M1

*All correct (no + c etc)*

A1

When  $x = -1$ , gradient =  $-4$

CSO

A1

3

(v) Gradient of normal =  $\frac{1}{4}$

B1ft

$y = \text{"their gradient"} (x \pm 1)$

*Must be finding **normal**, not tangent*

M1

$$y = \frac{1}{4}(x + 1)$$

CSO; any correct form eg  $4y - x = 1$

A1

3

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[18]

**Q3.**

(a) (i)  $\frac{dy}{dx} = 4 \times \frac{1}{2} x^{-\frac{1}{2}} - \frac{3}{2} x^{\frac{1}{2}} = 2x^{-\frac{1}{2}} - \frac{3}{2} x^{\frac{1}{2}}$

*A power decreased by 1*

M1

*A1 For each correct term*

A1 A1

3

(ii) At  $P(4, 0)$ ,  $\frac{dy}{dx} = \frac{2}{\sqrt{4}} - \frac{3}{2} \times 2$

Attempts  $\frac{dy}{dx}$  when  $x = 4$

M1

$$= 1 - 3 = -2$$

AG

A1

2

(iii) Gradient of normal =  $\frac{1}{2}$   
*Use of or stating  $m \times m' = -1$*

M1

Equation of normal is  $y - 0 = m [x - 4]$   
*m numerical; can be awarded even if  $m = -2$*

M1

$$y - 0 = \frac{1}{2}(x - 4) \Rightarrow 2y = x - 4$$

*ACF of the equation*

A1

3

(iv) At Q,  $x = 0$ ,  $2y = 0 - 4$   
*PI*

M1

$$y_e = -2$$

$$\text{Area of triangle } OPQ = 0.5 \times 4 \times |y_e|$$

*Ft on a linear equation for normal provided  $y_e$  is negative and prev A1 is lost*

A1F

$$= 4$$

*Ft on c's negative  $y_e$*

B1F

3

$$(v) \quad 2x^{-\frac{1}{2}} - \frac{3}{2}x^{\frac{1}{2}} = 0 \Rightarrow 2x^{-\frac{1}{2}} = \frac{3}{2}x^{\frac{1}{2}}$$

Puts  $c$ 's  $\frac{dy}{dx} = 0$  and a 1<sup>st</sup> step in attempt to solve.

M1

$$2 = \frac{3}{2}x \quad \Rightarrow x = \frac{4}{3}$$

Valid method to  $ax = b$

m1

Condone 1.3 or better

A1

3

$$(b) \quad (i) \quad \int \left( 4x^{\frac{1}{2}} - x^{\frac{3}{2}} \right) dx = 4 \frac{x^{\frac{3}{2}}}{1.5} - \frac{x^{\frac{5}{2}}}{2.5} \{+c\}$$

$$= \frac{8}{3}x^{\frac{3}{2}} - \frac{2}{5}x^{\frac{5}{2}} \{+c\}$$

One power correct

M1

Condone absence of "+ c"

A1,A1

3

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$$(ii) \quad \text{Area under curve} = 4 \frac{4^{\frac{3}{2}}}{1.5} - \frac{4^{\frac{5}{2}}}{2.5} - \{0\}$$

$$F(4) - \{F(0)\}$$

M1

Total area =  $F(4)$  + area triangle  $OPQ$

m1

$$\text{Total area} = \frac{128}{15} + 4 = \frac{188}{15} = 12.5(3...)$$

Accept 3 sf if clear

A1

3

[20]

**Q4.**

(a) (i)  $y = 2x^2 - 8x + 3$   
 $\left(\frac{dy}{dx}\right) = 4x - 8$

B1

1

(ii)  $\int_4^6 \frac{x-2}{2x^2-8x+3} dx$   
 $= \frac{1}{4} [\ln |2x^2 - 8x + 3|]_4^6$

*M1 for  $k \ln (2x^2 - 8x + 3)$ ; allow  $k \ln u$*

M1A1

$$= \frac{1}{4} [\ln 27 - \ln 3]$$

*Correct substitution into  
 $k \ln (2x^2 - 8x + 3)$  or 3, 27 into  $k \ln u$*

m1

$$= \frac{1}{4} \ln 9$$

$$= \frac{1}{2} \ln 3$$

A1

4

(b)  $\int x\sqrt{3x-1} dx$   
 $u = 3x - 1 \quad du = 3 dx$   
 OE

B1

$$\int = \left(\frac{1}{9}\right) \int \left(u^{\frac{3}{2}} + u^{\frac{1}{2}}\right) (du)$$

*[2 terms in with rational indices*

M1

$$= \left(\frac{1}{9}\right) \left[ \frac{u^{\frac{5}{2}}}{\frac{5}{2}} + \frac{u^{\frac{3}{2}}}{\frac{3}{2}} (+c) \right]$$

*Must be 2 terms with correct indices*

*(only fit for  $x = \frac{u-1}{3}$ )*

A1F

$$= \frac{2}{45} (3x-1)^{\frac{5}{2}} + \frac{2}{27} (3x-1)^{\frac{3}{2}} + c$$

CSO OE

A1

4

[9]

**Q5.**

(a)  $\text{Grad } AC = \frac{15}{3} = 5$   
OE

B1

Equation of AC:  $y = m(x + 2)$

or  $(y - 15) = m(x - 1)$

*Or use of  $y = mx + c$  with  $(-2, 0)$  or*

*$(1, 15)$  correctly substituted for  $x$  and  $y$*

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M1

$$y = 5x + 10$$

OE eg  $y - 15 = 5(x - 1)$ ,  $y = 5(x + 2)$

A1

3

(b) (i)  $\left[ 16x - \frac{x^5}{5} \right]$

*Raise one power by 1*

M1

*One term correct*

A1

*All correct*

A1

$$\left(16 - \frac{1}{5}\right) - \left(-32 + \frac{32}{5}\right)$$

$F(1) - F(-2)$  attempted

m1

$$= 41\frac{2}{5} \quad (\text{or } 41.4, \frac{207}{5} \text{ etc})$$

CSO; withhold if + c added

A1

5

$$(ii) \quad \text{Area } \Delta = \frac{1}{2} \times 3 \times 15 \quad \text{or } 22\frac{1}{2} \quad \text{or } 22.5$$

$$\text{Or } \int_{-2}^1 (5x + 10) dx = 22.5$$

B1

Shaded area = "their (b)(i) answer" – correct triangle

Condone "difference" if  $\Delta > \int$

M1

$$\Rightarrow \text{shaded area} = 18\frac{9}{10}$$

CSO; OE (18.9 etc)

A1

3

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[11]

**Q6.**

$$(a) \quad \left(1 + \frac{4}{x^2}\right)^3 = \left[1^3\right] + 3(1^2)\left(\frac{4}{x^2}\right) + 3(1)\left(\frac{4}{x^2}\right)^2 + \left[\left(\frac{4}{x^2}\right)^3\right]$$

Any valid method as far as term(s) in  $1/x^2$  and term(s) in  $1/x^4$

M1

$$= [1] + \frac{12}{x^2} + \frac{48}{x^4} + \left[\frac{64}{x^6}\right]$$

$$p = 12 \text{ Accept } \frac{12}{x^2} \text{ even within a series}$$

A1

$$q = 48 \text{ Accept } \frac{48}{x^4} \text{ even within a series}$$

A1

3

(b) (i)  $\int \left(1 + \frac{4}{x^2}\right)^3 dx = \int \left(1 + \frac{p}{x^2} + \frac{q}{x^4} + \frac{64}{x^5}\right) dx$

*Integral of an 'expansion', at least 3 terms  
PI by the next line*

M1

$$= x - px^{-1} - \frac{q}{3}x^{-3} - \frac{64}{5}x^{-5} (+c)$$

$$= x - 12x^{-1} - 16x^{-3} - \frac{64}{5}x^{-5} (+c)$$

*At least two powers correctly obtained*

m1

*Ft on c's non-zero integer values for p and q (A1F for two terms correct; can be unsimplified)  
Condone missing c but check that signs have been simplified at some stage before the award of both A marks.*

A2F,1

4

(ii)  $\left(2 - \frac{p}{2} - \frac{q}{3(8)} - \frac{64}{5(32)}\right) - \left(1 - p - \frac{q}{3} - \frac{64}{5}\right)$

*F(2) – F(1), where F(x) is cand's answer or the correct answer to (b)(i).*

M1

$$= 33.4$$

CSO

A1

2

[9]

**Q7.**

(a)  $\frac{dy}{dx} = -x^2 \sin 2x + \cos 2x$

*product rule  $kx \sin 2x \pm \cos 2x$*

M1

*no further incorrect working*

A1

2

(b) (i)  $-2\alpha \sin 2\alpha + \cos 2\alpha = 0$

*replacing  $x = \alpha$  and writing equation equal to zero (at any line)*

M1

$$\left. \begin{array}{l} 2\alpha \sin 2\alpha = \cos 2\alpha \\ 2\alpha \tan 2\alpha = 1 \end{array} \right\} \text{either}$$

$$2\alpha \tan 2\alpha - 1 = 0$$

AG; CSO

A1

2

(ii)  $\left. \begin{array}{l} f(0.4) = 0.2 \\ f(0.5) = -0.6 \end{array} \right\} \text{awrt o.e.}$   
*(0.9's unsubstantiated scores M0)*

M1

Change of sign  $\therefore 0.4 < \alpha < 0.5$

A1

2

(iii)  $2x \tan 2x = 1$

$$\left. \begin{array}{l} \tan 2x = \frac{1}{2x} \\ 2x = \tan^{-1}\left(\frac{1}{2x}\right) \end{array} \right\} \text{either}$$

$$x = \frac{1}{2} \tan^{-1}\left(\frac{1}{2x}\right)$$

AG; CSO

B1

1

(iv)  $x_1 = 0.4$

$$x_2 = 0.4480\dots$$

$$x_2 = 25.7$$

M1

$$x_3 = 0.4200... \\ = 0.42$$

A1

2

(c)  $y = x \cos 2x$

$$\left. \begin{array}{l} u = x \quad du = 1 \\ dv = \cos 2x \quad v = \frac{\sin 2x}{2} \end{array} \right\}$$

*differentiate one term*  
*integrate one term* } must be  $k \sin 2x$

M1

*correct substitution of their values into  
parts formula using  $u = x$*

m1

$$\int = \frac{x \sin 2x}{2} - \int \frac{\sin 2x}{4} (dx) \\ = \left[ \frac{x \sin 2x}{2} + \frac{\cos 2x}{4} \right]_{(0)}^{(0.5)}$$

A1

$$= \left( \frac{\sin 1}{4} + \frac{\cos 1}{4} \right) - \left( \frac{\cos 0}{4} \right)$$

*correctly substituting values from  
previous 2 method marks*

m1

$$= 0.0954$$

AWRT

A1

5

[14]

**Q8.**

(a) (i)  $B(0, 5)$

B1

$$\text{Area } AOB = \frac{1}{2} \times 1 \times 5$$

*Condone slip in number or a minus sign*

M1

$$= 2\frac{1}{2}$$

A1

3

(ii)  $\frac{3x^6}{6} + \frac{2x^2}{2} + 5x$  or  $\frac{x^6}{2} + x^2 + 5x$  (may have + c or not)

*Raise one power by 1*

M1

*One term correct*

A1

*All correct unsimplified*

A1

3

(iii) Area under curve =  $\int_{-1}^0 f(x) dx$   
*Correctly written or  $F(0) - F(-1)$  correct*

B1

$$[0] - \left[ \frac{1}{2} + 1 - 5 \right]$$

*Attempt to sub limit(s) of -1 (and 0)  
 Must have integrated*

M1

Area under curve  $3\frac{1}{2}$   
**CSO** (no fudging)

A1

Area of shaded region =  $3\frac{1}{2} - 2\frac{1}{2} = 1$   
**FT** their integral and triangle (very generous)

B1 ✓

4

(b) (i)  $\frac{dy}{dx} = 15x^4 + 2$

*One term correct*

M1

*All correct (no +c etc)*

A1

when  $x = -1$ , gradient = 17

**CSO**

A1

3

(ii)  $y = \text{"their gradient"} (x + 1)$

*Must be finding **tangent** – not normal  
any form e.g.  $y = 17x + 17$*

B1 ✓

1

[14]

**Q9.**

(a) (i)  $y = x + 1 + 4x^{-2} \Rightarrow \frac{dy}{dx} = 1 - 8x^{-3}$

M1

*Power  $p \rightarrow p - 1$*

*(A1 if  $1 + ax^n$  with  $a = -8$  or  $n = -3$ )*

A2,1,0

3

(ii)  $1 - 8x^{-3} = 0$

*Puts c's  $\frac{dy}{dx} = 0$*

M1

$x^3 = 8$

*Using  $x^{-k} = \frac{1}{x^k}$  to reach  $x^a = b$ ,  $a > 0$  or correct use of logs.*

m1

$x = 2$

A1

When  $x = 2$ ,  $y = 4$

A1ft

4

(iii) At (1, 6),  $\frac{dy}{dx} = 1 - 8 = -7$   
*Attempt to find  $y'(1)$*

M1

Gradient of normal =  $\frac{1}{7}$   
*Use of or stating  $m \times m' = -1$*

M1

Equation of normal is  $y - 6 = m[x - 1]$   
*m numerical*

M1

$$y - 6 = \frac{1}{7}(x - 1)$$

$$\left\{ \frac{y - 6}{x - 1} = \frac{1}{7}; 7y = x + 41 \right\}$$

*OE ft on c's answer for (a)(i) provided at least A1 given in (a)(i) and previous 3M marks awarded*

Alft

4

(b) (i)  $\int x \left( 1 + \frac{4}{x^2} \right) dx =$

$$\dots\dots\dots = \frac{x^2}{2} + x - 4x^{-1} \{+ c\}$$

*One of three terms correct.*

M1

*One of three terms correct.  
 For A2 need all three terms as printed or better  
 (A1 if 2 of 3 terms correct)*

A2,1,0

3

(ii) {Area = }  $\int_1^4 x + 1 + \frac{4}{x^2} dx =$

$$\left[ \frac{x^2}{2} + x - \frac{4}{x} \right]_1^4 = (8 + 4 - 1) - \left( \frac{1}{2} + 1 - 4 \right)$$

*Dealing correctly with limits;  $F(4) - F(1)$   
(must have integrated)*

M1

= 13.5

A1

2

[16]

**Q10.**

(a) (i)  $\int (4 - e^{2x}) dx$

=  $4x - \frac{1}{2} e^{2x} (+c)$

4x

-  $\frac{1}{2} e^{2x}$

B1

B1

2

(ii)  $\int_0^{\ln 2} = \left[ 4x - \frac{1}{2} e^{2x} \right]_0^{\ln 2}$

=  $\left[ 4 \ln 2 - \frac{1}{2} e^{2 \ln 2} \right] - \left[ (0) - \frac{1}{2} (e^0) \right]$

*Substitute both  $\ln 2$  and 0 correctly into  
an integrated expression*

M1

=  $4 \ln 2 - 2 + \frac{1}{2}$

*Convincing*

=  $4 \ln 2 - \frac{3}{2}$

**AG**

A1

2

(b) (i)  $x = 0$

$$y = 4 - 1 = \underline{\underline{3}}$$

B1

1

(ii) At  $B$ ,  $y = 0$

$$4 - e^{2x} = 0$$

*Or reverse argument*

M1

$$e^{2x} = 4$$

$$x = \ln 2$$

**AG**

A1

2

(c)  $\frac{dy}{dx} = -2e^{2x}$

B1

$$x = \ln 2, \text{ Gradient} = -2e^{2\ln 2} = -8$$

$$x = \ln 2 \text{ into } ke^{2x}$$

M1

$$\text{Gradient normal} = \frac{1}{8} = \frac{1}{2e^{2\ln 2}}$$

**OE**

A1

Equation  $y = -\frac{1}{8}x - \frac{1}{8}\ln 2$

**OE**

A1

4

(d) When  $x = 0$

$$y = -\frac{1}{8} \ln 2$$

*Attempt to integrate their line and substitute  
 $x = 0, \ln 2$*

M1

$$\text{Area } \Delta = \frac{1}{16} (\ln 2)^2 \quad \text{condone - ve sign} = 0.03$$

$$\frac{1}{2} (\text{their } y) \times \ln 2$$

A1 ✓

$$\text{Total area } 4 \ln 2 - \frac{3}{2} + \frac{1}{16} (\ln 2)^2 = 1.30$$

CSO                      AWRT

A1

3

[14]

**Q11.**

- (a) (i)  $f(1) = 1 + 4 - 5$   
*must find  $f(1)$  NOT long division*

M1

$$\Rightarrow f(1) = 0 \Rightarrow (x - 1) \text{ is factor}$$

*shown = 0 plus a statement*

A1

2

- (ii) Attempt at  $x^2 + x + 5$   
*long division leading to  $x^2 \pm x \dots$  or  
 equating coefficients*

M1

$$f(x) = (x - 1)(x^2 + x + 5)$$

$p = 1, q = 5$  by inspection scores B1, B1

A1

2

- (iii)  $(x =)1$  is real root

B1

Consider  $b^2 - 4ac$  for their  $x^2 + x + 5$   
*not the cubic!*

M1

$$b^2 - 4ac = 1^2 - 4 \times 5 = -19 < 0$$

Hence no real roots (or only real root is 1)  
*CSO; all values correct plus a statement*

A1

3

(b) (i)  $\int \dots dx = \frac{x^2}{4} + 2x^2 - 5x (+c)$   
*one term correct unsimplified*

M1

*second term correct unsimplified*

A1

*all correct unsimplified*

A1

3

(ii)  $[4 + 8 - 10] - \left[ \frac{1}{4} + 2 - 5 \right]$   
*correct use of limits 1 and 2;  
 F(2) - F(1) attempted*

M1

$$= 4 \frac{3}{4}$$

A1

Area of  $\Delta = \frac{1}{2} \times 11 = 5 \frac{1}{2}$

*correct unsimplified*

B1

$\Rightarrow$  shaded area  $= 5 \frac{1}{2} - 4 \frac{3}{4}$

*combined integral of  $7x - 6 - x^3$  scores  
 M1 for limits correctly used then*

$$= \frac{3}{4}$$

A3 correct answer with all working correct

A1

4

[14]

**Q12.**

(a) (i)  $x^2$

B1

1

(ii)  $x^{\frac{1}{2}} = \sqrt{x}$

Accept either form

B1

1

(iii)  $x^3$

B1

1

$$\int 3x^{\frac{1}{2}} dx = \frac{3}{\frac{3}{2}} x^{\frac{3}{2}} \{+c\}$$

(b) (i)

Index raised by 1

M1

Simplification not yet required

A1

$$= 2x^{\frac{3}{2}} + c$$

Need simplification and the + c OE

A1

3

(ii)  $\int_1^9 3x^{\frac{1}{2}} dx = (2 \times 9^{\frac{3}{2}}) - (2 \times 1^{\frac{3}{2}})$

$F(9) - F(1)$ , where  $F(x)$  is candidate's answer to (b)(i) [or clear recovery]

M1

$$= 52$$

*Ft on (b)(i) answer of form  $kx^{1.5}$  i.e.  $26k$*

A1ft

2

[8]

**Q13.**

(a)  $\int x e^{5x} dx$

$$u = x \quad dv = e^{5x}$$

*integrate one term, differentiate one term*

M1

$$du = 1 \quad v = \frac{1}{5} e^{5x}$$

A1

$$\int = \frac{1}{5} x e^{5x} - \int \frac{1}{5} e^{5x} dx$$

A1

$$= \frac{1}{5} x e^{5x} - \frac{1}{25} e^{5x} (+c)$$

A1

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(b) (i)  $u = x^{\frac{1}{2}}$

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$$du = \frac{1}{2} x^{-\frac{1}{2}} dx$$

M1

$$\int = \int \frac{1}{1+u} \times 2 du$$

*correct with no errors; AG*

A1

2

(ii)  $\int_1^9 dx = \int_1^3 \frac{2}{1+u} du$

*correct limits used in correct expression, ignoring k*

m1

$$= [2 \ln(1+u)]_1^3$$

*for k ln(1+u)*

M1

$$= 2 \ln 4 - 2 \ln 2$$

$$(\ln 4)$$

*ISW OE*

A1

3

[9]

**Q14.**

(a)  $\tan x (+ c)$

B1

1

(b)  $f(x) = \frac{\cos x}{\sin x}$

$$f'(x) = \frac{-\sin^2 x - \cos^2 x}{\sin^2 x}$$

*quotient rule*  $\frac{\pm \sin^2 x \pm \cos^2 x}{\sin^2 x}$

M1

A1

$$= \frac{-1}{\sin^2 x}$$

*use of  $\sin^2 x + \cos^2 x = 1$*

A1

$$= -\operatorname{cosec}^2 x$$

*AG CSO*

A1

4

**Special cases**

$$f(x) = \frac{\cot x}{1}$$

$$f'(x) = \frac{1 \times -\operatorname{cosec}^2 x - \cot x \times 0}{1^2}$$

M1

$$= -\operatorname{cosec}^2 x$$

A1

(max 2/4)

**Or**

$$f(x) = \frac{1}{\tan x}$$

$$f'(x) = \frac{\tan x \times 0 - 1 \times \sec^2 x}{\tan^2 x}$$

M1A1

$$= \frac{-\sec^2 x}{\tan^2 x}$$

$$= \frac{-1}{\sin^2 x} = -\operatorname{cosec}^2$$

A1

(max 3/4)

(c) LHS =  $\tan^2 x + \cot^2 x + 2 \tan x \cot x$   
expanding

M1

$$= \tan^2 x + 1 + \cot^2 x + 1$$

correct use of trig identities

M1

$$= \sec^2 x + \operatorname{cosec}^2 x$$

CSO

A1

3

$$= \text{RHS}$$

(d)  $\int (\tan x + \cot x)^2 dx = \int \sec^2 x + \operatorname{cosec}^2 x dx$

use of identity

M1

$$= [\tan x - \cot x]_{0.5}^1$$

$$\pm \tan x \pm \cot x \text{ OE}$$

M1

A1

$$= 0.9153 - -1.2842$$

$$= 2.2$$

AWRT

A1

4

[12]

**Q15.**

(a)  $y_D = 3 + 1 = 4$  or  $y_C = 12 - 8 = 4$

*Attempt at either y coordinate*

M1

$$\text{Area } ABCD = 3 \times 4 = 12$$

A1

2

(b) (i)  $x^3 - \frac{x^4}{4} \quad (+C)$

*Increase one power by 1*

M1

*One term correct unsimplified*

A1

*All correct unsimplified (condone no +C)*

A1

3

(ii) Sub limits -1 and 2 into their (b)(i) ans

*May use both -1, 0 and 0, 2 instead*

M1

$$[8 - 4] - \left[ -1 - \frac{1}{4} \right] = 5 \frac{1}{4}$$

A1

Shaded area = "their" (rectangle- integral)

*Alt method: difference of two integrals*

M1

$$= 12 - 5 \frac{1}{4} = 6 \frac{3}{4}$$

**CSO.** Attempted M2, A2

A1

4

(c) (i)  $\frac{dy}{dx} = 6x - 3x^2$

*One term correct*

M1

*All correct ( no +C etc)*

A1

2

(ii) When  $x = 1, y = 2$  when  $x = 1,$

*May be implied by correct tgt equation*

B1

$$\frac{dy}{dx} = 3 \text{ as 'their' grad of tgt}$$

*Ft their derivative when  $x = 1$*

M1ft

Tangent is  $y - 2 = 3(x - 1)$

*Any correct form  $y = 3x - 1$  etc*

A1

3

(iii) Decreasing when  $\frac{dy}{dx} = 6x - 3x^2 < 0$

*Watch no fudging here!!*

*May work backwards in proof.*

M1

$$3(2x - x^2) < 0 \Rightarrow x^2 - 2x > 0$$

**AG** (be convinced no step incorrect)

A1

2

(d) Two critical points 0 and 2

*Marked on diagram or in solution*

M1

$x > 2$ ,  $x < 0$  ONLY  
 or M1 A0 for  $0 < x < 2$  or  $0 > x > 2$   
 SC B1 for  $x > 2$  (or  $x < 0$ )

A1

2

[18]

**Q16.**

(a)  $\frac{dy}{dx} = \frac{3}{2}x^{\frac{1}{2}} - 3$

One term correct

M1

Both correct

A1

2

(b) (i) When  $x = 0$ ,  $\frac{dy}{dx} = -3$   
 Ft provided answer  $< 0$ .

B1F ✓

Eqn of tangent at  $O$  is  $y = -3x$   
 OE Ft on  $y'(0)$

B1F ✓

2

(ii) At  $(9,0)$   $\frac{dy}{dx} = \frac{3}{2}(9)^{\frac{1}{2}} - 3$

Attempt to find  $y'(9)$

M1

Eqn tangent at  $A$  is  $y - 0 = y'(9)[x - 9]$   
 OE

m1

$\Rightarrow y = \frac{3}{2}(x - 9) \Rightarrow 2y = 3x - 27$   
 CSO. AG

A1

3

(iii) Eliminating  $y \Rightarrow -6x = 3x - 27$

*OE method to one variable*

(eg  $2y = -y - 27$ )

M1

$$9x = 27 \Rightarrow x = 3$$

*[A1F for each coordinate; only ft on  $y = kx$  tangent in (b)(i) for  $k < 0$ ]*

A1F

When  $x = 3, y = -9$ .  $P(3, -9)$

A1F

3

(c)  $\int \left( x^{\frac{3}{2}} - 3x \right) dx = \frac{2}{5} x^{\frac{5}{2}} - \frac{3x^2}{2} (+c)$

*One power correct*

M1

*Condone absence of "+ c" and unsimplified forms*

A2,1,0

3

(d)  $\int_0^9 \left( x^{\frac{3}{2}} - 3x \right) dx =$   
PI

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B1

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$$= \frac{2}{5} \times 9^{\frac{5}{2}} - \frac{3}{2} \times 9^2 - 0$$

*Correct use of limits following integration*

M1

$$= -24.3$$

$$\text{Area of triangle } OPA = \frac{1}{2} \times 9 \times |y_P|$$

M1

$$\text{Sh. Area} = \frac{1}{2} \times 9 \times |y_P| - \left| \int_0^9 \left( x^{\frac{3}{2}} - 3x \right) dx \right|$$

OE

M1

$$= 40.5 - 24.3 = 16.2$$

A1

5

[18]

**Q17.**

(a) (i)  $f' = \frac{dy}{dx} = 4x^3 + 2$

B1

1

(ii)  $\int \frac{2x^3 + 1}{x^4 + 2x} dx$

$$= \frac{1}{2} \ln(x^4 + 2x) + c$$

*For k ln (x<sup>4</sup> + 2x)*

*By substitution k ln u*    M1  
*correct*    A1

M1

A1

2

(b) (i)  $u = 2x + 1$

$$du = 2 dx$$

B1

$$\int x\sqrt{2x+1} dx =$$

$$\int \left(\frac{u-1}{2}\right)\sqrt{u} \frac{du}{2}$$

*Must be in terms of u only incl. du*

M1

$$= \frac{1}{4} \int \left(u^{\frac{3}{2}} - u^{\frac{1}{2}}\right) du$$

AG

A1

3

$$(ii) \int_0^4 dx = \int_1^9 du$$

*Or changing u's to x's at end*

B1

$$\frac{1}{4} \int u^{\frac{3}{2}} - u^{\frac{1}{2}} = \frac{1}{4} \left[ \frac{u^{\frac{5}{2}}}{\frac{5}{2}} - \frac{u^{\frac{3}{2}}}{\frac{3}{2}} \right]$$

M1

A1

$$= \frac{1}{4} \left[ \left( \frac{2}{5}(9)^{\frac{5}{2}} - \frac{2}{3}(9)^{\frac{3}{2}} \right) - \left( \frac{2}{5} - \frac{2}{3} \right) \right]$$

$$= \frac{1}{4} [79.2 + 0.26]$$

$$= 19.86$$

*Sight of any of these 3 lines*

$$= 19.9$$

AG

A1

4

EXAM PAPERS PRACTICE

[10]

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**Q18.**

$$(a) y = x^{-2} \ln x$$

$$\frac{dy}{dx} = x^{-2} \frac{1}{x} - 2x^{-3} \ln x$$

*Use of product or quotient*

M1

*each term*

A1 A1

$$= \frac{1 - 2 \ln x}{x^3}$$

*Convincing argument*  $x^{-2} \times \frac{1}{x} = x^{-3}$   
 AG

A1

4

(b)  $\int x^{-2} \ln x \, dx$       $u = \ln x$       $dv = x^{-2}$   
*Attempt at integration by parts*

M1

$$du = \frac{1}{x} \quad v = -x^{-1}$$

A1

$$\int = -\frac{1}{x} \ln x + \int x^{-2} \, dx$$

A1

$$= -\frac{1}{x} \ln x - \frac{1}{x} (+C)$$

A1

4

(c) (i) At A,  $\frac{dy}{dx} = 0$

$$1 - 2 \ln x = 0$$

$$\ln x = \frac{1}{2}$$

*Attempt at  $\ln x = k$*

M1

$$x = e^{\frac{1}{2}}$$

A1

2

$$(ii) \quad R = \left[ -\frac{1}{x} (\ln x + 1) \right]_1^5$$

$$R = [\text{Their (b)}]_1^5$$

M1

$$= -\frac{1}{5} (\ln 5 + 1) + (\ln 1 + 1)$$

OE

A1

$$= \frac{1}{5} (4 - \ln 5)$$

*convincing argument; AG*

A1

3

[13]

**Q19.**

(a)  $\cos 2x = 2 \cos^2 x - 1$

B1B1

2

(b)  $\cos^2 x = \frac{1}{2} (\cos 2x + 1)$

*Attempt to express  $\cos^2 x$  in terms of  $\cos 2x$*

M1

$$\frac{1}{2} \int_0^{\frac{\pi}{2}} \cos 2x + 1 \, dx = \left[ \frac{1}{4} \sin 2x + \frac{x}{2} \right]_0^{\frac{\pi}{2}}$$

A1

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A1

$$= \frac{\pi}{4}$$

*Use limits. Ft on integer a.*

M1A1F

5

[7]

**Q20.**

(a) (i)  $\frac{dy}{dx} = 3x^2 - 20x + 28$

*One term correct*

M1

*Another term correct*

A1

*All correct (no + c etc)*

A1

3

- (ii) Their  $\frac{dy}{dx} = 0$  for stationary point  
*Or realising condition for stationary point*

M1

$$(x - 2)(3x - 14) = 0$$

*Attempt to solve using formula/ factorise*

m1

$$\Rightarrow x = 2$$

A1

$$\text{or } x = \frac{14}{3}$$

*Award M1, A1 for verification that  $x = 2 \Rightarrow \frac{dy}{dx} = 0$   
 then may earn m1 later*

A1

4

(b) (i)  $\frac{x^4}{4} - \frac{10x^3}{3} + 14x^2 (+ c)$

*One term correct unsimplified*

M1

*Another term correct unsimplified*

A1

*All correct unsimplified (condone missing + c)*

A1

3

(ii)  $\left[ \frac{81}{4} - 90 + 126 \right] \quad (-0)$

*Attempt to sub limit 3 into their (b)(i)*

M1

$$= 56\frac{1}{4}$$

*AG Integration, limit subtraction all correct*

A1

2

(iii) Area of triangle =  $31\frac{1}{2}$

*Correct unsimplified  $\frac{1}{2} \times 21 \times 3$*

B1

Shaded Area =  $56\frac{1}{4} - \text{triangle area}$

M1

$$= 24\frac{3}{4}$$

*Or equivalent such as  $\frac{99}{4}$*

B1

3

[15]

Q21.

EXAM PAPERS PRACTICE

(a)  $y = x \ln x$  ©2025 Exam Papers Practice. All rights reserved.

$$\frac{dy}{dx} = x \times \frac{1}{x} + \ln x$$

*use of product rule (only differentiating, 2 terms with + sign)*

M1

$$= \ln x + 1$$

A1

2

(b)  $\int (\ln x + 1) dx = x \ln x$

*OE; attempt at parts with*

M1

$$\int \ln x \, dx = x \ln x - x (+c)$$

A1

2

(c)  $\int_1^5 \ln x \, dx = [x \ln x - x]_1^5$

$$= (5 \ln 5 - 5) - (1 \ln 1 - 1)$$

*correct substitution of limits into their (b) provided  $\ln x$  is involved*

M1

$$5 \ln 5 - 4$$

*ISW*

A1

2

[6]

**Q22.**

- (a) (i)  $x - 1$  is a factor

*May be earned retrospectively*

B1

$x + 2$  is a factor

*From part (iii)*

B1

2

- (ii) Attempt at third factor

*Multiplying/ dividing/factor theorem*

M1

$$f(x) = (x - 1)(x + 2)(x - 4)$$

$$(x+4) \Rightarrow M1, A0$$

A1

2

- (b) (i) At  $A$ ,  $y = 8$

*Or (0,8)*

B1  
1

- (ii) At  $B$ ,  $x = 4$   
*Or (4,0) NO ft of wrong factor*

B1  
1

- (c) (i)  $\frac{x^4}{4} - x^3 - 3x^2 + 8x (+c)$   
*Increase one power by 1*

M1

*One term correct (unsimplified)*

A1

*Two other terms correct (unsimplified)*

A1

*All correct (unsimplified)  
 (condone missing + c)*

A1  
4

- (ii) Realisation that limits are  $-2$  and  $1$   
*Condone wrong way round*

B1

$$\text{Area} = \left[ \frac{1}{4} - 1 - 3 + 8 \right] - [4 + 8 - 12 - 6]$$

*Attempt to sub their limits into their (c)(i)*

M1

$$= 20 \frac{1}{4}$$

*CSO. Must use  $F(1) - F(-2)$  correctly*

A1  
3

[13]

**Q23.**

(a)  $\sqrt{x} = x^{\frac{1}{2}}$

*Accept  $k = 0.5$*

B1

1

(b)  $\sqrt{x}(x-1) = x^{\frac{1}{2}}x - x^{\frac{1}{2}} = x^{\frac{3}{2}} - x^{\frac{1}{2}}$

M1

Accept  $p = 1.5, q = 0.5$

A1

2

(c)  $\int \sqrt{x}(x-1) dx = \frac{x^{2.5}}{2.5} - \frac{x^{1.5}}{1.5} (+c)$

Increases a power of  $x$  by 1

M1

ft non-integer  $p$

A1ft

ft non-integer  $q$

A1ft

3

(d)  $\int_1^2 dx = \left( \frac{2^{2.5}}{2.5} - \frac{2^{1.5}}{1.5} \right) - \left( \frac{1}{2.5} - \frac{1}{1.5} \right)$

Limits;  $F(2) - F(1)$

M1

... =  $\left( \frac{4\sqrt{2}}{2.5} - \frac{2\sqrt{2}}{1.5} \right) - \left( \frac{1}{2.5} - \frac{1}{1.5} \right)$

Fractional powers to surds

m1

$\left( \frac{24\sqrt{2}}{15} - \frac{20\sqrt{2}}{15} \right) - \left( -\frac{4}{15} \right) = \text{pr. ans}$

CSO AG (be convinced)

A1

3

[9]

**Q24.**

- (a) (i)  $p(3) = 27 - 45 + 21 - 3$   
*Finding  $p(3)$*

M1

$p(3) = 0 \Rightarrow x - 3$  is a factor  
*Shown = 0 plus a statement*  
 Or  $(x - 3)(x^2 - 2x + 1)$

A1

2

- (ii)  $B$  is point  $(3,0)$   
*Must have coordinates*

B1

1

- (b) (i)  $\frac{dy}{dx} = 3x^2 - 10x + 7$   
*One term correct*

M1

*All correct with NO + c etc*

A1

2

- (ii)  $3x^2 - 10x + 7 = 0$

*Putting their*  $\frac{dy}{dx} = 0$

M1

$\Rightarrow (x - 1)(3x - 7) = 0$   
*Attempt to use quad formula or factorise*

m1

$\Rightarrow$  at  $M$ ,  $x = \frac{7}{3}$   
*CSO factors correct etc*

A1

3

- (c)  $\frac{d^2y}{dx^2} = 6x - 10$

ft their  $\frac{dy}{dx}$

B1ft

sub  $x = 1, \Rightarrow \frac{d^2y}{dx^2} = -4$

ft their  $\frac{d^2y}{dx^2}$

B1ft

2

(d) (i)  $\frac{x^4}{4} - \frac{5x^3}{3} + \frac{7x^2}{2} - 3x$  (+c)  
*Increase one power by 1*

M1

*One term correct*

A1

*Two other terms correct*

A1

*All correct (condone missing + c)*

A1

4

- (ii) Realisation that limits are 0 and 1  
*Condone wrong way round*

B1

$$\left[ \frac{1}{4} - \frac{5}{3} + \frac{7}{2} - 3 \right] - 0$$

*Attempt to sub their limits into their (d)(i)*

M1

$$= -\frac{11}{12}$$

**CSO.** Must use  $F(1) - F(0)$  correctly

A1

$$\text{Area} = \frac{11}{12}$$

**CSO. Convincing argument**

E1

4

[18]

**Q25.**

(a) (i)  $\sqrt{x} = x^{\frac{1}{2}}$   
 Accept  $p = 0.5$

B1

1

(ii)  $\int \sqrt{x} \, dx = \frac{x^{1.5}}{1.5} \{+c\}$   
 Index raised by 1

M1

Correct ft on p. Condone missing '+c'

A1ft

2

(iii) Area =  $\int_1^4 \sqrt{x} \, dx$   
 Limits 1 and 4 PI

B1

..... =  $\frac{4^{1.5}}{1.5} - \frac{1}{1.5}$   
 $F(4) - F(1)$

M1

=  $\frac{14}{3}$

Accept 4.66 or better

A1

3

(b) (i)  $y = x^{\frac{1}{2}} \Rightarrow \frac{dy}{dx} = \frac{1}{2} x^{-\frac{1}{2}}$   
 Index (p - 1) ft

M1

When  $x = 4$ ,  $y'(4) = 0.25$

*Attempt to find  $y'(4)$*

M1

When  $x = 4$ ,  $y = 2$

B1

Equation of tangent:  $y - 2 = \frac{1}{4}(x - 4)$

*accept other forms*

A1

4

(ii) When  $x = 0$ ,  $y = 1$   $B(0, 1)$

*Subs  $x = 0$  **and** then  $y = 0$  into equation of tangent. PI*

M1

When  $y = 0$ ,  $x = -4$   $A(-4, 0)$

*Correct ft  $y_B$  and  $x_A$   
(may be awarded as part of area calculation)*

A1ft

Area =  $0.5(1)(4) = 2$

*ft wrong sloping tangent and max of 1 further slip. Final answer must be +ve*

A1ft

3

(c) Translation

*'Translation'/'translate(d)'*

B1

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

*Accept equivalent in words provided linked to 'translation/move/shift'  
(Note: B0B1 is possible)*

B1

2

(d)  $h = 1$

*PI*

B1

$\text{Integral} = h/2 \{ \dots \}$   
 $\{ \dots \} = f(1) + 2[f(2) + f(3)] + f(4)$   
*OE summing of areas of the three traps*

M1

$\{ \dots \} = 0 + 2(1 + \sqrt{2}) + \sqrt{3}$   
*Condone 1 numerical slip*

A1

$\text{Integral} = \frac{1}{2} \{ 2(1 + 1.414\dots) + 1.732 \}$   
 $\text{Integral} = 0.5 \times 6.560\dots = 3.28 \text{ to 3sf}$   
*Must be 3.28*

A1CAO

4

[19]

**Q26.**

(a) Stretch (parallel) to  $x$ -axis

B1

Scale factor  $\frac{1}{2}$

Translate  $\begin{pmatrix} 0 \\ 3 \end{pmatrix}$

B1

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B1,B1

4

(b)  $A = \int e^{2x} + 3 \, dx$   
*(+ attempt to integrate)*

M1

$= \left[ \frac{1}{2} e^{2x} + 3x \right]$   
*(correct)*

A1

$$\left(\frac{1}{2}e^8 + 12\right) - \left(\frac{1}{2}e^4 + 6\right)$$

*Substitute 2,4 into their [*

m1

$$\frac{1}{2}(e^8 - e^4) + 6$$

$$\left(\frac{1}{2}e^4(e^4 - 1) + 6\right)$$

A1

4

(c)  $x_1 = 2, \quad y_1 = e^4 + 3 \quad (57.6)$

*Attempt at y(2) or y(4)*

M1

$x_2 = 4, \quad y_2 = e^8 + 3 \quad (2980)$

*Both correct*

A1

Area of  $A + B =$

$$2(e^8 - e^4) + 2(e^8 + 3)$$

*Attempt to find correct area*

M1

Area  $B =$   
 $4e^8 - 2e^4 + 6$

$$- \frac{1}{2}e^8 + \frac{1}{2}e^4 - 6$$

$$= \frac{7}{2}e^8 - \frac{3}{2}e^4$$

A1

4

[12]

**Q27.**

(a)  $\sin 2x = 2 \sin x \cos x$

B1

1

(b) (i)  $A = B = x$

M1

$$\cos 2x = \cos^2 x - \sin^2 x$$

A1

2

(ii)  $\cos (2x + x) = \cos 2x \cos x - \sin 2x \sin x$

M1

$$= (\cos^2 x - \sin^2 x) \cos x - 2 \sin^2 x \cos x$$

*ft (a) and (b)(i)*

A1ft

$$= \cos^3 x - 3 (1 - \cos^2 x) \cos x$$

*Eliminate all sines*

M1

$$= 4 \cos^3 x - 3 \cos x$$

*AG; convincingly obtained*

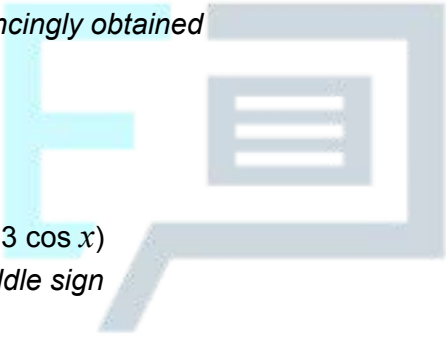
A1

4

(c)  $\cos^3 x = \frac{1}{4} (\cos 3x + 3 \cos x)$

*Ignore middle sign*

M1

  
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$$\int_0^{\frac{\pi}{2}} \cos^3 x dx = \frac{1}{4} \left[ \frac{1}{3} \sin 3x + 3 \sin x \right]_0^{\frac{\pi}{2}}$$

*1 for integrating each term*

A1A1

*Use of limits*

M1

$$\frac{1}{4} \left( -\frac{1}{3} + 3 \right) = \frac{2}{3}$$

AG

A1

5

[12]